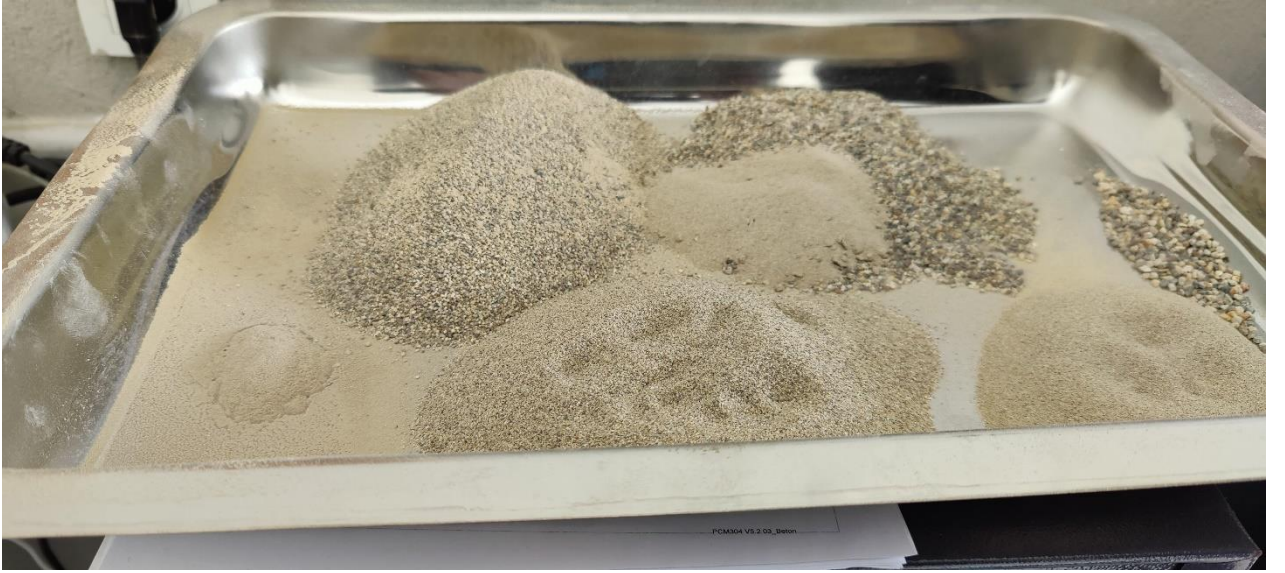


RAW MATERIALS



MAY 2023

SAND

The sand is the first material necessary for a good quality concrete roof tile, it is necessary to choose a sand or possible sands to be supplied to the factory but that allows to achieve some initial characteristics on the fines of the sand and that its granulometry, in its mixtures, achieves an "optimal curve of prefixed granulometry".

Most of the factories in their storage hoppers are prepared for a possible mixture of 2 or 3 sands, which is usual in the installations to achieve a good mixture of sand.

In reference to the characteristics of the type of sand to be used, it is preferable that the sand be river sand because of the better cohesion achieved in the mixture (sand, pigment cement, water) and the easier extrusion of the concrete tile, with less wear on the components of the extruder (rollers and slippers).

It is possible to use crushed sand, in the case of using it is advisable in my experience with a maximum percentage of 15% of the total, a percentage higher than that amount as 60% can be used, but the mixer will require a greater demand / efficiency in the moisture control of the final mixture and ensure a good mixing, in order to achieve a stable moisture content of the mixture to achieve greater fluidity in the extruder, it will also cause a greater/rapid wear of the rollers and slippers, an increase in the electrical consumption of the roller and the pusher chain, due to the greater pressure suffered by the slippers during the extrusion of the tile.

The following is an example of a possible real mixture of sands in a factory with the need to use 3 sands due to the initial requirements on the sand fines and the pre-set granulometry.

REFERENCE VALUES FOR SAND FINES:

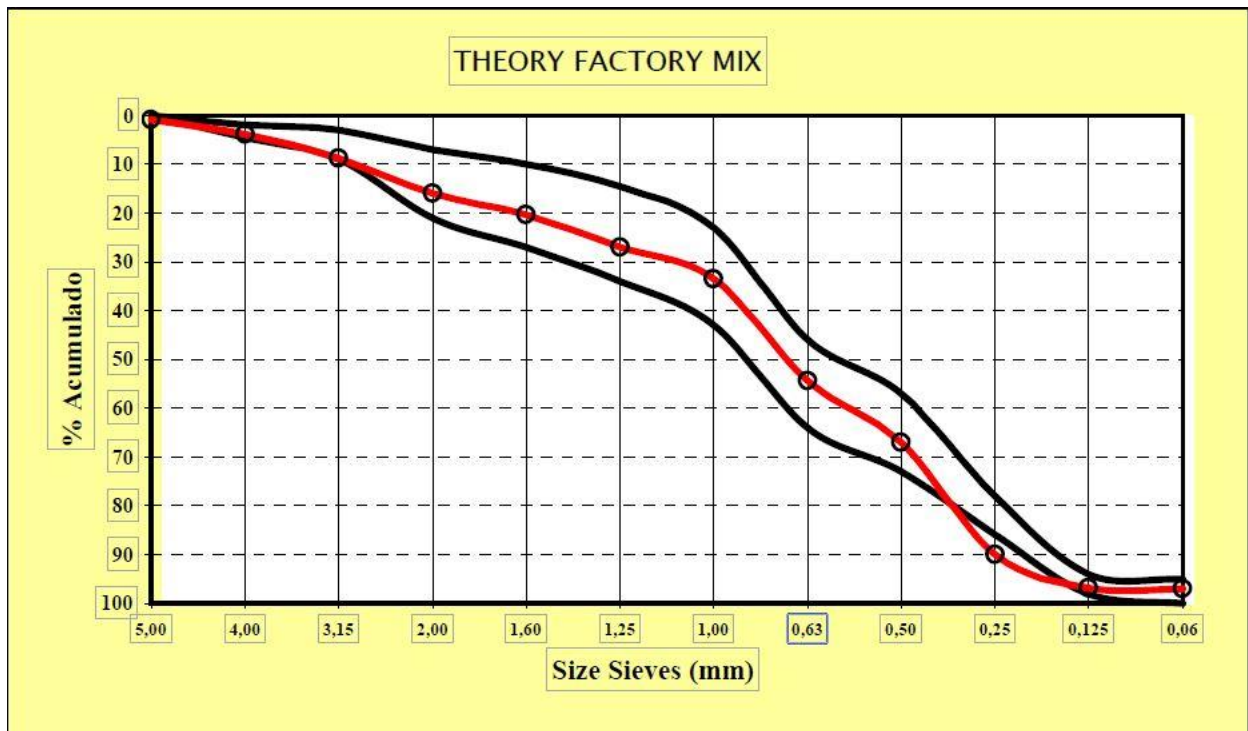
Fineness Modul (2,7 % to 3 %)

Fines 1 (2 % to 5 %)

Sand Equivalent (80 % al 85 %)

Theory Mix	
SAND A 0/4	77%
CRUSHING SAND. 0/4	6%
SAND E 0/1	17%

CONCRETE TILES		CONTROL AND ANALYSIS SAND																	
Sand Tests		GRANULOMETRY													WASHED		OTROS		
Sands	Sieve (mm)	5,00	4,00	3,20	2,00	1,60	1,25	1,00	0,63	0,50	0,25	0,125	0,06	Polvo	MF	Finos	E.Are.		M.Org
SAND A 0/4	%Retenido	1.1	3.7	5.6	8.3	5.3	7.8	7.6	20.3	10.1	19.9	7.6	0.1	2.7	3.19	1.7	83.7		NO
	%Acum.	1.1	4.7	10.4	18.7	24.0	31.7	39.4	59.6	69.7	89.6	97.2	97.3	100.0					
CRUSHING SAND. 0/4	%Retenido	0.8	3.7	8.8	13.3	6.4	7.2	5.2	10.0	4.7	11.4	12.2	0.3	16.1	2.91	11.2	67.2		NO
	%Acum.	0.8	4.5	13.3	26.6	32.9	40.1	45.3	55.3	60.0	71.4	83.6	83.9	100.0					
SAND E 0/1	%Retenido					0.1	0.5	2.3	27.1	27.0	41.2	1.7		0.0	2.58	0.10	99.1		NO
	%Acum.					0.1	0.6	2.9	30.0	57.0	98.2	100.0	100.0	100.0					
Theory Factory Mix		%Retenido	0.87	3.05	4.87	7.19	4.46	6.50	6.57	20.83	12.61	22.92	6.90	0.13	3.02	3.07	1.97	85.30	
	%Acum.	0.87	3.92	8.79	15.98	20.44	26.94	33.52	54.35	66.96	89.94	96.84	96.98	100.00					



The particle size we are looking to achieve with the mixture, according to experience, is that it is as close as possible to the particle size values in the sand indicated below.

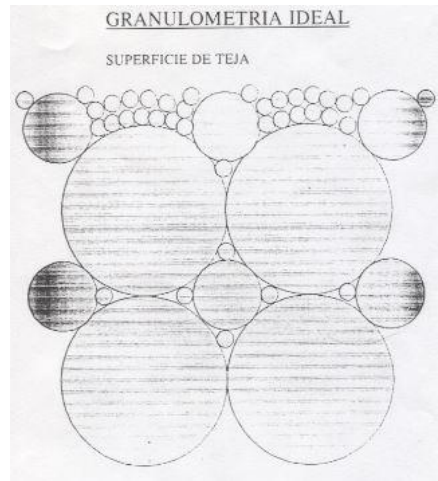
nº Sieve.	Size in	mm.	% Acumul	% o total
	4	4,76	0	0
	8	2,38	10	10
	12	1,7	17	7
	116	1,19	27	10
	30	0,59	57	30
	50	0,297	78	21
	100	0,149	94	16
	200	0,074	97	3
Powder		0,05	100	3
				100

It should be noted that the tile with more fine sand has a better surface appearance and increases the weight, the more coarse sand increases the strength, but worsens the surface appearance of the tile and reduces the weight on the concrete tile, all this with "the same thickness in all parts of the tile", by changing its density.

Regarding the minerals in the sand it is important to ensure that the sand used for the manufacture of the concrete roof tile does not contain high levels of sulphates (cracking and fissuring), chlorides (durability), organic matter (voids and bubbles in the tile) or soluble salts (efflorescence).

It may be necessary to use special additives or replace the sand with a more suitable type to ensure the quality and durability of the concrete roof tile.

IDEAL SAND



COVERED ENVIRONMENT SEPARATORS



INTERIOR FACTORY A COVERED.



SCREEN

The use of a Screen is essential to avoid the entrance of stones in the sand that can take the suppliers in the supplies of sand with the truck or that for some reason they are in the sand storage of the factory.

The entrance of a stone greater than the minimum thickness of production in the tile can break the mold and what is worse the roller or sleeper of conformation.

It is absolutely necessary if for the manufacturing it is going to use equipment of rollers and sleepers of widia.

About the type of Screens to use, in each factory they have to find which is the one that meets their efficiency and maintenance requirements from their own experience, in my opinion after years using different types of equipment I have my decision on which is the best Screen with guarantees and reduction of maintenance costs, the only and big problem is its initial cost.



“ HEIN, LEHMANN “ System : Liwell

<https://www.youtube.com/watch?v=xLZwFj4qeR8>

TIPE LSF 1,0 2,58/8 ED

Support legnth: 2.700 mm

Distance between silentblocks: 2.600 mm

Width between silentblocks: 1.200 mm

SCREEN



CEMENT

Types of cement supplier

The cement normally used is PORTLAND

According to Certifications **I 42,5R PORTLAND.**

I 52,5R PORTLAND

Normally concrete tiles are produced with Portland 42.5 and/or Portland 52.5, depending on the countries (North / South) and the need for more or less curing hours, so that they can have a good demoulding (without breakage) and with sufficient strength and less brittleness to allow good packaging and palletising.

According to the characteristics of the cement manufacturer in Spain, I 42,5 PORTLAND has a setting time of 60 minutes and I 52,5 Portland has a setting time of 45 minutes, which means that the hardening time of I 52,5 Portland is faster, but with enough time to be able to produce the tile without problems in the extruder head.

I 52.5 Portland also achieves a higher strength of 50 % in 48 hours compared to I 42.5 Portland.

The time it takes to reach its final strength or cure is 28 days,

The mission of the cement is to react with the sand and the water, to provoke the setting of the resulting material with the help of the curing process of the tiles in their chambers, to obtain a tile with a strength that allows its demoulding (depalletiser) and its transfer by the polycorers and its subsequent packaging and palletising.

The color cement needs to be monitored if the tile is not painted, since with only powder pigment the colour of the final tile can be lighter or darker”

What is the ideal amount of cement for the tile?

There is a theoretical approximation of the kilos per mix, but in the end it will be determined by the practical results of the resistance tests in the factory.

On the other hand, when there is a double shift and it is possible mixed (I 42,5 Portland + I 52,5 Portland) with another more reactive cement, the proportions will be determined in the situation of the tiles in the demoulding and packaging.

Real examples :

1 Shift 350 kg. I 42,5 Portland

2 Shift 380 kg. (280 kilos I 42,5 Portland + 70 kg. I 52,5 Portland)

The ratio used of sand / cement Spain **3,64.**

RATIO SAND / CEMENT

The ratio used of sand / cement most used in in the different profiles of the factories I have visited:

3,05 to 4,09

The ratio used of sand / cement Spain.... 3,64.

The ratio used of sand / cement Italy..... 3,30.

VAULTS IN CEMENT WEIGHING HOPPERS

It is convenient to replace the pneumatic / electric vibrators with impact vibrators to break the vaults created inside the cement weigh hoppers.



PIGMENT

Powder Pigment is a very standardised product, so today there are many suppliers of this synthetic iron oxide.

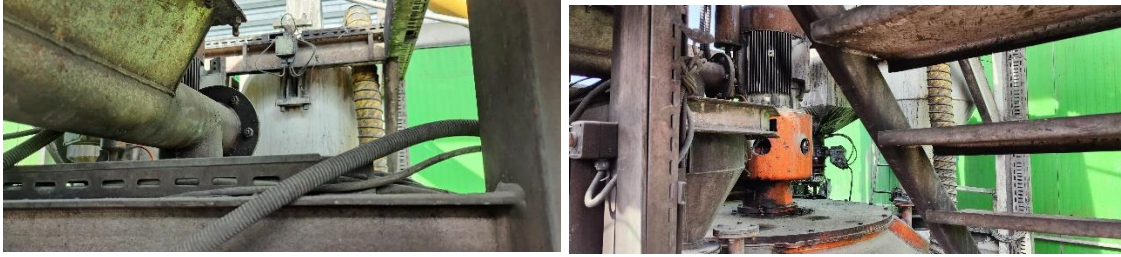
BAYER AG, CATHAY PIGMENTS, LANXESS, ROCKWOOD ITALIA (SILO)

MANUAL - DOSAGE OF POWDER PIGMENT TO THE MIXER



AUTOMATIC- DOSAGE OF POWDWER PIGMENT TO THE MIXER





Always from the beginning of the factory has worked with powder pigment manually.

The pigment was supplied in paper bags and more later in big-bag.

Depending on the accuracy required, the pigment is added manually by volume.

The best thing is a table near the mixer and a weighing scale with some cubes to make the recipe.

*As a reference for pigment weights, we can indicate actual weighing of pigments in different factories with different proportions and possibilities of pigment for each tile color. (Without surface painting)

*In order to match the tile colors of all the factories, we must compensate different color of the sands and cement. (Without surface painting)

P / C : Grey 2,8 % to 3,18 % .

P / C : Brown 2,7 % to 3,64 % .

P / C : Grey 4,55 % (Flat tile by color intensity).

LIQUID PIGMENT

Most of the factories in the world use powder pigments, liquid pigments are in the minority, I personally always recommend powder pigments.

The installation of the Liquid Pigment takes up a lot of space and they have to be in constant agitation to avoid decanting.

I can improve the dispersion, but I do not like the use of more water because of the pigment.

It is possible to realize by means of big-bag of pigment emptying it in a big container and rest water, keeping it always in constant agitation

ADDITIVES.

CHEMICAL PRODUCTS

Many years ago, we added cement, fly ash, steel slag, by-products, etc.

The cement companies were added to their cement recipe.

PLASTICIZING

DO NOT use chemical products, powders or liquids in the tile mix that can alter the structural characteristics (to achieve a greater initial resistance or a greater plasticity of the mortar) (accelerators, stiffeners, setting retarders, etc.) so far in my experience I have not identified that provide more advantages, on the contrary if I have identified disadvantages in its use. (There may be other criteria in this regard, which I respect but do not share).

In factories for surface finishes such as slurry, small doses are used for the formation of a 1 mm layer and alter its characteristics.

WATERPROOFING

Add elements / components for a contribution of fines (if the sand used did not have enough fines) whose objective was to increase and solve problems of impermeability without other purpose and not to alter the structure of the resulting concrete.

I consider it necessary to add a little waterproofing to guarantee waterproofing.

I can only contrast the use of calcium stearate and zinc powder has been my experience in this product because we had a supplier and a good price.

In additions to the manual mixer were made by volume.

With values of 0.822 grams / tile to a maximum (exceptional) of 4 grams.

The one that requires more waterproofing by profiles, is the flat tile.

The waterproofing tests for the guarantee certificate give the evaluation of the waterproofing addition needs,

ACCELERATOR ADDITIVES

This additive I do not use, nor have I ever used, and the least recommended for my part, as it does not guarantee the quality properties of the tiles that we demand in the Production Process in Spain, until they personally prove me otherwise in a factory.

Obtaining a high strength in a concrete tile in a short time has always been an objective for the technicians of all Central Laboratories.

The additive accelerator in the production has not been recommended because it has a great reduction of the setting time that can cause failures in the humidity control sensor of the mixer, difficulties in the shaping of the tile in the extruder box, subsequent problems in the formation of micro-cracks, fissures and cracks in the curing stage in the chambers, the quality defects are very difficult to identify when the tile is painted on its surface.

WATER FEEDING MIXER.

NATURAL WATER / COLD

Feed water to the mixer by well water or the external network



RECOVERED WATER FROM DECANTATION POND.



HOT WATER

Use hot water in the mixer to get the 20° to 25° temperature of the concrete in the mixer where it is necessary. (Boiler / Exchanger)



STEAM WATER

The purpose of adding steam to the mixer is to reach a mixing temperature of 20 to 25°, normally in the winter season, and to break up any grains of frozen sand from the yard or from the roof tiles, in order to achieve a good mix in the mixer, as they could cause a defect on the surface of the tile.